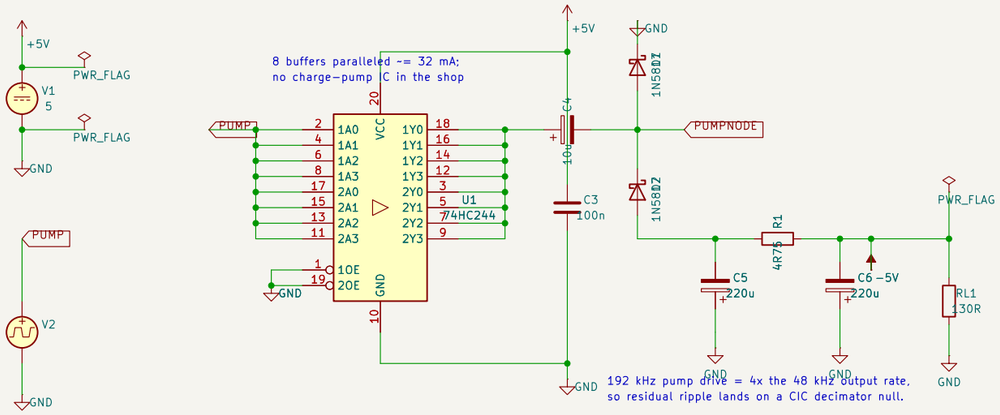
A The −5 V charge pump (74HC244 x8 paralleled, 192 kHz, loaded at 32 mA)



PUMP is 4040 Q4 = 192 kHz, which is 4x the 48 kHz output rate on purpose: whatever ripple survives lands exactly on a null of the CIC decimator. sim_c_clock makes it for real.

130R is the analog section: two TL074, a TL072 and the DAC offset legs, about 32 mA. That current is the whole reason the comparators run on a single +5 V supply -- leaving them on the negative rail costs 8 mA and the TL074s then cannot clear their 1.94 V peak swing.

Measured: −3.87 V at 29.8 mA, settling in about 40 ms, with 0.24 uV of ripple after 4R75 + 220u. The design's "under 1 uV" holds; its −4.1 V does not.
The pump's output resistance is about 16 ohm and it is set by the 74HC244's own on-resistance, not by 1/fC: 10 uF through 4 ohm is a 40 us time constant against a 2.6 us half period, so the flying cap never finishes charging. A bigger C4 or a faster PUMP will not fix that; only more parallel buffers would -- and there are no more.
The 40 ms matters at bring-up: measure the rail straight after power-on and it is still climbing.

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